

PREPARATION , CHARACTERISATION AND ENGINE PERFORMANCE EVALUATION WITH BLENDS OF BIODIESEL USING WASTE COOKING OIL



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ABSTRACT

The rapid increase of vehicle density all over the world has resulted in the upsurge of fuel consumption & depletion of fossil fuels. The exorbitant rates of fuel need no special mention. The adverse effects of the conventional fossil fuels and rise in fuel price have made researchers to carry out their research on various sources of Alternate Fuels, Biodiesel being one of them.

The process of producing biodiesel from vegetable oil is not strictly economical due to cost of the raw materials although there are reductions in emission gases from automobile exhaust. Waste cooking oil on the other hand is not eco-friendly and difficult to dispose. This oil can be used as a partial substitute for diesel to run internal combustion engine to economize the fuel cost, reduce environmental pollution, and ameliorate the difficulties of unburnt cooking oil disposal. This study envisages an attempt to convert the waste cooking oil (palm oil) into Biodiesel using the process of transesterification and use it as a supplement to diesel. An internal combustion engine will be run with different blends (specifically 5%, 10% and 15%) by substituting Diesel with Biodiesel produced from cooking oil waste. The properties like Brake Power(BP), Brake specific fuel consumption (BSFC), Fuel consumption(FC), Brake Thermal efficiency and exhaust gas temperature will be measured.

1.1 Biofuels:

In the context of shrinking crude oil reserves, rising demand and the resultant rise in prices of petroleum, as well as the concerns about global climate change and energy security. Bio energy is becoming increasingly relevant as a possible and potential alternative to fossil Fuels. Biofuels are liquid or gaseous fuels produced from biomass resources and used in place of, or in addition to diesel, petrol or other fossil fuels for transport, stationary, portable and other applications. Biofuels are derived from renewable bio-mass resources and, therefore, provide a strategic advantage to promote sustainable development and to supplement conventional energy sources in meeting the rapidly increasing requirements for transportation fuels associated with high economic growth.

There are four types of biofuels:

1.1.1 First generation biofuels: First generation biofuels are made from biomass consisting of sugars, starch, vegetable oils, animal starch or biodegradable output wastes from agriculture, industry, forestry and households using conventional technologies. Biodiesel and bioethanol come under first generation biofuels [7].

1.1.2 Second generation biofuels: Second generation bio-fuel technologies are gaining importance because manufacturing first generation biofuels have major limitations. The primary one is that they cannot be produced beyond a threshold level without threatening food security. They are also not cost-competitive with existing fossil fuels. The second-generation fuels are more sustainable, affordable and have greater environmental benefits. However, they have not become popular because the technology for producing these is not yet standardized. Further, Lignocellulosic materials can be converted to alcohol. Organic waste material converted into energy forms which can be used as an automotive fuel: waste oil (e.g. cooking oil) into biodiesel [21].

1.1.3 Third and Fourth generation biofuels: The third-generation biofuels include Algal biodiesel, Algal hydrogen and conversion of biomass to hydrogen whereas the fourth-generation biofuels include biofuels from high solar efficiency cultivation. However, these are still at a nascent stage. Biofuels offer a number of environmental, social and economic advantages. The use of biofuels may lead to reduction in vehicular pollution and greenhouse gas emissions as it is established that the emission of sulphur dioxide (SO₂), particulate matter and carbon monoxide (CO) etc. are less from biofuels [7]

Our present fuel resources are not going to be around forever and with the ever-increasing consumption their extinction is nearly unavoidable. Fuel resources which are mostly made up of fossil fuels are not renewable in nature.

Currently around the world the consumption of fossil fuels is 100,000 times faster than their natural production. According to an estimate the demand for these fuels will suddenly outstrip their availability in a matter of centuries or less. Another fact which should also be paid attention to here is that the consumption is eating up our environment very fast.

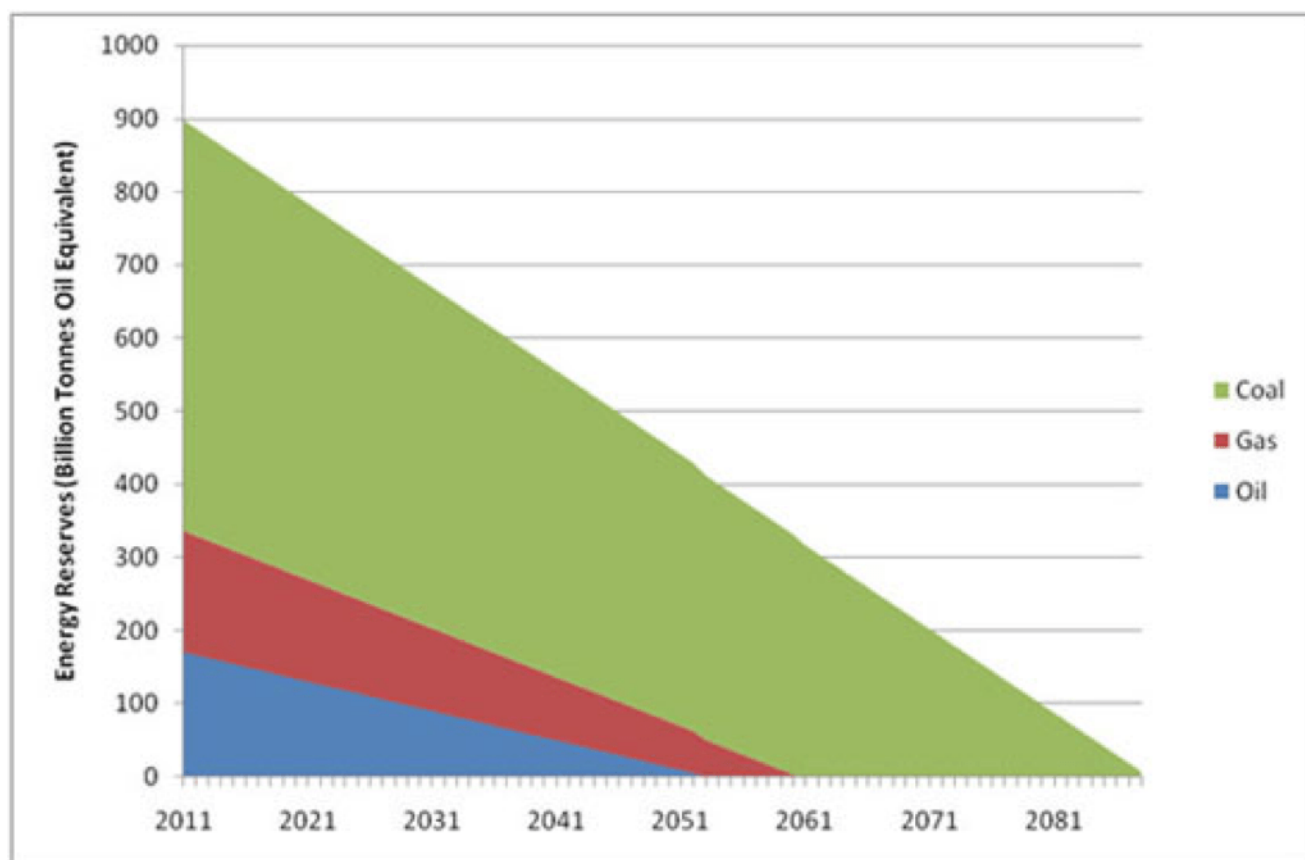


Fig. 1: End of Fossil Fuel Graph

No matter how safely the fossil fuels are used, they continue to take a toll on the atmosphere. This is because on combustion these types of fuels release various pollutants. These pollutants include dangerous gases like sulphur dioxide, carbon dioxide etc.

The rampant warnings of global warming are the biggest reason for the world to look beyond these fossil fuels. The consumption is not the only thing that adds to the pollution. The development in technology has made it possible to extract fossil fuels in a much easier and more cost-effective way. This involves digging deeper into the ground, which causes scraping away of ever more layers of precious topsoil. Alternative fuels on the other hand are safer. They do not cause pollution in the atmosphere, nor do they cause the depletion of natural resources. Also, unlike the fossil fuels, the resources for the alternative fuels are entirely renewable.

Biodiesel is a renewable fuel that can be produced from vegetable oils, animal fats, used cooking oil, and waste from the pulp and paper industry. It can be used in its neat form, or as a blend with conventional diesel fuel, in diesel engines without any modifications. Since biodiesel is produced from renewable, domestically grown feed stock, it can reduce the use of petroleum-based fuels and possibly lower the overall greenhouse gas (carbon dioxide, methane, nitrous oxide & ozone) emissions from the use of internal combustion engines.

Table 1: Properties of other alternate fuels compared to Bio-Diesel

	LPG	Natural Gas (CNG)	Hydrogen	Methanol	Bio-Diesel
Density	0.5- 0.58kg/dm ³	0.656kg/m ³	0.08375kg/m ³	792kg/m ³	874kg/m ³
Viscosity	5.38x10 ⁻⁶ lb/fts	0.01-0.03cp	1.04x10 ⁻⁵ Pa s	0.6906x10 ⁻⁶ m ² /s	3.6-5.0 cSt
Cetane Number	NA	NA	NA	NA	48-65
Octane Number	90-112	135	130+	112	NA
CV of Fuel (KJ/kg)	55000	50000	150000	22700	37270
Ignition Temperature	470 °C	1004 °F	1050-1080°F	470°C	406°C
Flash Point	-104 °C	-300°F	NA	54°F	100-170°C
Fire Point	410-580 °C	540°C	500°C	470°C	700°C
Boiling Point	-42 °C	-4°C	-252.9°C	64.7°C	315-350°C
Freeze Point	-188 °C	-162°C	-259.14°C	97.6°C	-15°C
Flame Temperature	1967 °C	700°C	2127°C	1870°C	800°C
Compression Ratio	11-13	16	17:1-18:1	14:1	17.5:1-15.5:1

1.2 Sources of Biodiesel

Biodiesel can be produced from a wide variety of oil seed crops and animal fats. In Europe, rapeseed oil is the major biodiesel feed stock. In the United States, soybeans are the dominant biodiesel feedstock.

Soybeans

Soybeans produce approximately 1.5 gallons of oil per bushel. U.S. farmers produced a record-breaking yield in 2009 of 44 bushels per acre. This translates to an oil yield of 66 gallons per acre. Soybean oil is a co-product with soybean meals, a popular high-protein feed for animals. Although soybeans do not produce as much oil per acre as other crops, such as canola or rapeseed, soy oil is the most popular oil in the United States. It is commonly grown as a rotation crop with corn. The United States has an established infrastructure to process soybeans into oil and meal.

Rapeseed and Canola

Rapeseed and canola produce about 75 to 240 gallons of oil per acre. They are excellent rotation crops because they have deep root systems that scavenge well for water and nutrients. These crops use planting and harvesting equipment similar to what is used for small-grain production, and the meal commands high prices in the feed market.

Mustard

Mustard, a relative of canola and rapeseed, has also been shown to be an excellent cover crop with high potential as a biodiesel feed stock. Although it produces less oil than canola, it is drought tolerant, grows well on marginal soils, and contains compounds that act as natural fumigants against soil pathogens.

Safflower and Sunflower

Safflowers and sunflowers both produce oil that can be used for biodiesel, although these oils tend to be more highly prized as premium cooking oils.

Olive Oil

Olives are a species of small tree in the family Oleaceae. The species is cultivated in many regions and considered naturalized in France, Crimea, Egypt, Syria, Java, Norfolk Island California, Bermuda etc. The fruit is a small drupe 1-2.5 cm (0.39-0.98 in) long, thinner-fleshed and smaller in wild plants than in orchard cultivars. Olives are harvested in the green to purple stage.

Olive trees show marked preference for calcareous soil, flourishing best a on limestone slopes and crags, and coastal climate conditions. They grow in any light soil, even on clay if well drained, but in rich soils they are predisposed to disease and produce poorer oil. Olives like hot weather and sunny positions without any shade while temperatures below -10 °C (14°F) may injure even a mature tree. Olive trees can live for several centuries and can remain productive for as long if they are pruned correctly and regularly.

Jatropha

A 114-page report published in 2010 by the Food and Agriculture Organization of the United Nations emphasizes that the versatility of *Jatropha* may make it useful in poor, remote areas of the world. This article summarizes the FAO report.

Castor Bean

A major constraint for using castor oil as a feed stock for biodiesel is the high price castor oil commands as an industrial and pharmaceutical feed stock. Castor oil boasts high lubricity characteristics, so biodiesel derived from castor oil could achieve the required lubricity for biodiesel standards at concentrations much lower than that of rapeseed (*Brassica napus* L.) or soybean (*Glycine max* L.). However, castor oil's high viscosity may limit its use to lower percentages in biodiesel blends or to warm climates.

Animal Fats

Animal fats are low-cost biodiesel feed stocks. However, there are some drawbacks and challenges to animal fat feed stocks.

Algae

Micro algae have long been recognized as potentially good sources for bio fuel production because of their relatively high oil content and rapid biomass production. There are production challenges which need to be overcome for successful commercialization.

Used and waste cooking oil

Waste oil (Example. **Palm oil**) and grease can be collected from restaurants to produce biodiesel. While the processing costs of this urban source are higher per gallon than the processing costs of virgin vegetable oil, the cost of the feed stock is generally low and sometimes free.

Table 2: Properties of Fossil Fuels and Vegetable oils

	Cetane/Octane Number	Specific Gravity (Kg/m³)	Viscosity (cP)	Calorific Value (MJ/Kg)	Flash Point (°C)	Sulphur Content (Wt. %)/ppm
Diesel	45-55	0.83	0.89	50	62	0.247
Petrol	87	0.79	0.71	45	43	0.014
Mustard Oil	45-55	0.953	24.67	32.43	254	2.4[ppm]
Soybean Oil	37.6	0.914	39.5	39.60	69	>15[ppm]
Jatropha Oil	33.7	0.916	35.98	38.8	280	-
Olive Oil	65	0.911	8.4	37	324	-
Waste cooking oil	52	0.893	4.318	39.76	156	0.0114

Table 3: Properties of Vegetable Oil based Biodiesel

	Soybean	Sunflower	Castor	Jatropha	Mustard	Palm
Kinematic Viscosity at 40°C(mm²/s)	32.6	33.9	251	34	42.77	39.4
Density (kg/l)	0.914	0.9161	0.960	0.94	0.913	0.919
Flash Point (°C)	254	274	229	225	218	252
Cloud Point (°C)	-4	-9.5	-18	11	-2	25.2
Iodine Number	138-143	125-140	83-86	82-98	101	48-58

1.3 Scenario of Waste cooking oil in India:

Waste cooking oil as raw material for biodiesel: Waste cooking oil consists of the remains of frying food composed of plant or animal fats with cooking oil. Cooking oil is a glycerol ester composed of different essential fatty acids that can only be dissolved in organic solvents. It is harmful to the environment and to consumer's health when cooking oil waste is consumed. When fried meals are utilized, it is believed that the poisonous substances generated when the oil oxidizes can cause cancer. Waste cooking Oil is widely available, although it is often abandoned after usage, particularly after frying. The **Food Safety and Standards Authority of India(fssai)** estimates that the amount of waste cooking oil produced in India is around **3 million metric tons** annually. The majority of spent cooking oil from houses is disposed of in the drainage and soil or returns to the food chain. Thus, one recommendation to overcome these issues is to utilize the waste cooking oil as raw materials for biodiesel which is more economical and environmentally friendly. It is also having potential to be a feed stock for biodiesel production as the amount of waste cooking oil produced will increase. The chemical reactions that occur during the frying process, which are oxidation, hydrolysis, polymerization, and material transfer between the food and the oil, cause the properties of waste cooking oil to differ from fresh oil. However, waste cooking oil has identical properties to produce biodiesel in terms of density, kinematic viscosity, acid value and iodine value and meet the quality requirement. Waste cooking oil contains a higher concentration of free fatty acids (FFA) than vegetable oil, which is significant when selecting a catalyst for use in the manufacturing process. In situations where the concentration of FFA is high, large-scale biodiesel production is restricted, while soap manufacturing is favored.

Based on previous studies, the values of kinematic viscosity, saponification, flash point, moisture content and free fatty acids in waste cooking oil are very high.

These chemical properties will influence the quality and amount production of biodiesel. Furthermore, it is difficult to inject fuel from fuel tank to engine if the concentration or viscosity of oil is high and will cause atomization to occur. Waste cooking oil is 2 to 3 times cheaper than fresh vegetable oil, resulting in a significant reduction in total processing costs. The retail price of biodiesel is significantly influenced by the cost of raw material as up to 75–90% of the entire biodiesel production cost is spent on procuring the necessary raw materials alone. Using waste cooking oil as a raw material avoids controversies such as the Food vs Fuel debate and related environmental concerns.

Table 4 :Prices of Edible oil in India (Values in Rs/Kg)

	2019	2020	2021	2022
Palm	76	90	140	160
Vanaspati	70	100	140	170
Mustard	100	120	160	200
Soybean	85	90	155	190
Sunflower	100	110	145	192
Groundnut	140	143	180	190

1.4 Advantages and Disadvantages of Biodiesel:

Advantages:

1. Biodiesel fuel is a renewable energy source unlike petroleum-based diesel.
2. Biodiesel reduces our dependence on oil from foreign countries and contributes to our own society.
3. Biodiesel is plant-based and adds almost no CO₂ to the atmosphere.
4. The lack of sulphur in 100% biodiesel extends the life of catalytic converters.
5. Another of the advantages of biodiesel fuel is that it can also be blended with other energy resources and oil.
6. Biodiesel has a higher flashpoint than conventional diesel.
7. Biodiesel is less toxic than conventional diesel.
8. Pleasant smelling, both in its raw form, and when burned.
9. Higher cetane rating.
10. Biodiesel can reduce the classic diesel engine "knocking" noise.
11. Burns up to 90% cleaner than conventional diesel fuel made from fossil fuel.

12. The ozone forming potential of biodiesel emissions is nearly 50% less than conventional diesel.

Disadvantages

1. It can harm rubber hoses in some engines.
2. Biodiesel fuel distribution infrastructure needs improvement, which is another of the biodiesel fuel disadvantages.
3. It cannot be transported in pipelines.
4. Biodiesel is less suitable for use in low temperatures.
5. It tends to reduce fuel economy.
6. It takes a large expanse of area to grow and they may raise the price of certain foods.

1.5 Selection of Oil:

From the above-mentioned sources of bio diesel, we have selected **Waste Palm oil** as our subject of interest.

- India is the world's second largest consumer of **Palm oil** and also the world's largest importer (10 million tonnes imported in the year 2018). Due to the low price of palm oil, it is popular among India's poorer citizens for household cooking.
- In 2021, the Indian government launched the National Mission on Edible Oils to promote domestic oil palm cultivation, focusing on the northeast region and the Andaman and Nicobar Islands. The goal was to reduce dependence on edible oil imports significantly.
- Due to the high availability and less cost, palm oil is used widely in restaurants, cafes and is a popular choice among local vendors. As a result, a large amount of waste palm oil is generated every day.

So we decided to recycle this waste oil and produce biodiesel out of it which will be cheaper than conventional fuel and also beneficial for the environment.

Table 5 : The State-wise and year-wise production of Crude Palm Oil in India during the year 2016-2021
(Values in Metric Tons)

Sl. No	State	2016-17	2017-18	2018-19	2019-20	2020-21
1	Andhra Pradesh (AP)	190999	234696	232938	208359	240016
2	Goa	436	379	411	309	305
3	Karnataka	2051	2224	2280	2184	2734
4	Kerala	5989	5191	4857	4825	4281
5	Mizoram	603	648	625	535	675
6	Tamil Nadu	1194	938	1017	532	698
7	Telangana	8947	27274	37205	38050	39347
	All India	210219	271349	279332	254794	288056

CHAPTER 2

LITERATURE REVIEW

Experiments and research have been conducted by various researchers regarding the production of biodiesel using various alternate fuels. Few of them are mainly focused on the production of bio diesel and analysis of its chemical properties. Others emphasize the emissions created when biodiesel is blended with diesel and used in an IC engine.

Patni Neha have analyzed of current scenario of Biofuels in India specifically Biodiesel and Bioethanol. They stated that an up-to-date technology policy is central to bring in efficiency in production which is also cost-effective so that the industry would survive on its own without any subsidies or support and the focus on research has to be sustained to explore the feasibility of environmentally friendly and economically sustainable feed stocks.

Piyanuch Nakpong and Sasiwimol Wootthikanokkhan (2009) investigated in the production of biodiesel from three mixtures of vegetable oil and used cooking oil by alkali catalysed transesterification. Three kinds of vegetable oils, including Jatropha, roselle and coconut oils were tested. The effect of used cooking oil content in oil feed stock (used cooking oil/vegetable oil ratios of 0.03-0.2 v/v) on methyl ester formation was investigated and optimized. The methyl ester content from each reaction condition was determined by gas chromatography (GC). The optimum used cooking oil/vegetable oil ratio was 0.03 v/v for all three kinds of oil feed stock. At this ratio, the methyl ester content of three kinds of biodiesel was higher than the minimum limit set for Thai biodiesel (B100) specification. The measured properties of three biodiesel products were within the limit prescribed by Thai standards, except for the lower viscosity of biodiesel produced from coconut-used cooking oil mixture. However, it was very close to that of Thai petroleum diesel.

S.L.Sinha and R.K.Yadav (2010) investigated, the biodiesel produced from the Jatropha seeds have been considered as a potential alternative for running the compression ignition engines. The different blends of biodiesel and conventional diesel have been tested on the engine. The experimental data obtained for various concentrations of biodiesel blends have been compared with base line data of conventional diesel. Significant improvement in engine performance has been observed due to the use of biodiesel. Acceptable thermal efficiencies of the engine have been obtained with different blends of biodiesel and diesel. It has been observed that 20% of Jatropha oil can be substituted for diesel without any engine modification and preheating of the blends. The level of hydrocarbon emission and noise level have been found to be reduced with the use of more biodiesel content.

Hamed M. El-Mashad et al. (2008) investigated Salmon oil, a by-product of salmon processing, was used as a feed stock for biodiesel production via transesterification in a two-step process. Two different types of salmon oil were tested: salmon oil extracted from acidified salmon hydrolysate and salmon oil extracted from salmon by-products. Optimal amounts of chemicals required to give the highest biodiesel yield from each oil were determined using batch production procedures. It was found that due to the high acid value of salmon oil, alkaline catalysed transesterification was not an effective method for producing biodiesel from the salmon oil. Therefore, a two-step process was applied, in which a sulfuric acid-catalysed pre-treatment was used in the first step to reduce the acid value from 12.0 to 3mg [KOH] g [oil]⁻¹ and then, in the second step, KOH-catalyzed transesterification was applied. All experiments were performed at a temperature of 5272 1C with a mixing intensity of 600 rpm. Based on the total weight of salmon oil used, the maximum biodiesel yield of 99% was achieved using a total methanol/molar ratio of 9.2% and 0.5% (w/w) KOH. Ester loss due to the formation of emulsion during the washing and drying

steps was 15% maximum. This loss could be reduced in practical applications by better design of washing and drying techniques. A preliminary economic analysis showed that the cost of biodiesel production from salmon oil was almost twice that produced from soybean oil.

P.K. Sahoo et. al.(2006) results on non-edible filtered high viscous (72 cSt at 40 °C) and high acid value (44 mg KOH/gm) polanga (*Calophyllum inophyllum* L.) oil based mono esters (biodiesel) produced by triple stage transesterification process and blended with high speed diesel (HSD) were tested for their use as a substitute fuel of diesel in a single cylinder diesel engine. HSD and polanga oil methyl ester (POME) fuel blends (20%, 40%, 60%, 80%, and 100%) were used for conducting the short-term engine performance tests at varying loads (0%, 20%, 40%, 60%, 80%, and 100%). Tests were carried out over entire range of engine operations at varying conditions of speed and load. The brake specific fuel consumption (BSFC) and brake thermal efficiency (BTE) were calculated from the recorded data. The engine performance parameters such as fuel consumption, thermal efficiency, exhaust gas temperature and exhaust emissions (CO, CO₂, HC, NO_x, and O₂) were recorded. The optimum engine operating condition based on lower brake specific fuel consumption and higher brake thermal efficiency was observed at 100% load for neat biodiesel. From emission point of view the neat POME was found to be the best fuel as it showed lesser exhaust emission as compared to HSD.

Daniela Russi (2008) Biofuels are often presented as a contribution towards the solution of the problems related to our strong dependency on fossil fuels, i.e. greenhouse effect, energy dependency, urban pollution, besides being a way to support rural development. In this paper, an integrated assessment approach is employed to discuss the social desirability of a large-scale biodiesel production in Italy, considering social, environmental and economic factors. The conclusion is that the advantages in terms of reduction of greenhouse gas emissions, energy dependency and urban pollution would be very modest. The small benefits would not be enough to offset the huge costs in terms of land requirement: if the target of the European Directive 2003/30/EC were reached (5.75% of the energy used for transport by 2010) the equivalent of about one-third of the Italian agricultural land would be needed. The consequences would be a 14-considerable increase in food imports and large environmental impacts in the agricultural phase. Also, since biodiesel must be de-taxed in order to make it competitive with oil-derived diesel, the Italian energy revenues would be reduced. In the end, rural development remains the only sound reason to promote biodiesel, but even for these objective other strategies look more advisable, like supporting organic agriculture.

G Lakshmi Narayana Rao ,S Sampath, and K Rajagopal (2008) Trans esterified vegetable oils (biodiesel) are promising alternative fuel for diesel engines. Used vegetable oils are disposed from restaurants in large quantities. But higher viscosity restricts their direct use in diesel engines. In this study, used cooking oil was dehydrated and then trans esterified using an alkaline catalyst. The combustion, performance and emission characteristics of Used Cooking oil Methyl Ester (UCME) and its blends with diesel oil are analyzed in a direct injection C.I. engine. The fuel properties and the combustion characteristics of UCME are found to be similar to those of diesel. A minor decrease in thermal efficiency with significant improvement in reduction of particulates, carbon monoxide and unburnt hydrocarbons is observed compared to diesel. The use of trans esterified used cooking oil and its blends as fuel for diesel engines will reduce dependence on fossil fuels and also decrease considerably the environmental pollution.

N.R. Banapurmatha A, P.G. Tewaria A, R.S. Hosmath B (2008) described the best way to use vegetable oils as fuel in compression ignition (CI) engines is to convert it into biodiesel. Biodiesel is a methyl or ethyl ester of fatty acids made from vegetable oils (both edible and non-edible) and animal fat. It can be used in CI engines with very little or no engine modifications. This is because it has properties similar to mineral diesel. They present the results of investigations carried out on a single-cylinder, four stroke, direct-injection, CI engine operated with methyl esters of Honge oil, Jatropa oil and sesame oil.

Comparative measures of brake thermal efficiency, smoke opacity, HC, CO, NOX, ignition delay, combustion duration and heat release rates have been presented and discussed. Engine performance in terms of higher brake thermal efficiency and lower emissions (HC, CO, NOX) with sesame oil methyl ester operation was observed compared to methyl esters of *Honge* and *Jatropha* oil operation.

Michael S. Graboski et al. (1999) important part of this study, data on emissions based on the EPA heavy-duty transient cycle have been measured to demonstrate the sensitivity of engine emissions to biodiesel ester composition. To understand the impact of biodiesel composition on emissions, the testing included a series of fuels consisting of nearly pure fatty acid esters. A detailed analysis of the data was performed in an effort to determine what fuel properties are correlated with the increase in NO_x emissions observed for biodiesel here, and in most previous studies. In total, 27 neat biodiesels and 3 B-20 blends (with EPA certification diesel) were tested. Seven fuels prepared from various natural feedstocks were obtained from IGT, and three of these were also tested as B-20 blends. Certification fuel runs exhibited a mean NO_x emission level of 4.59 g/bhp-h (+/-0.3%) with no significant drift over the 4-month test program. PM emissions averaged 0.261 g/bhp-h (+/-2.7%). A small downward drift in PM emissions was evident (11% over the course of the project). Thus, no correction for drift was applied to the NO_x emissions data but a correction was applied to the PM data. Regression analysis of the results indicated that emissions could be correlated by one parameter, density of cetane number (which were highly correlated with each other). PM reduction was proportional to oxygen content for biodiesels with a cetane number of greater than about 45 (density greater than 0.89). For fuels with cetane number less than 45 PM reduction was less. NO_x emissions were also well correlated with biodiesel density or cetane number. These results suggest that neat biodiesels with cetane numbers greater than about 60 may produce NO_x emissions equal to or less than certification fuel.

V P Sethi, K S Salariya (2004) research on a 4-stroke 5 hp diesel engine was tested with two different fuel blends. In the first case, diesel kerosene blends (with 10% to 40% kerosene blending by volume) and in the second case, air-liquefied petroleum gas (LPG) mixture (15% to 25 % LPG mixing by volume) along with diesel was tested at constant engine speed of 1700 rpm. Different engine exhaust emissions, namely, carbon dioxide (CO₂), carbon monoxide (CO), unburnt hydrocarbons (UHC), sulfur dioxide (SO₂), oxides of nitrogen (NO_x) and unused oxygen (O₂) were compared using pure diesel, diesel-kerosene blends and air-LPG mixtures. With diesel-kerosene blends minimum exhaust emissions were observed at 30% kerosene blend. Exhaust gas emissions, namely CO, UHC, and SO₂ reduced by 40%, 18% and 19%, respectively, when compared with pure diesel emissions. Slight increase in the NO_x exhaust emission (2.4%) was observed. With air-LPG mixtures, minimum exhaust emissions were observed at 20% LPG mixing. Exhaust gas emissions, namely, CO, UHC, and SO₂ reduced by 80%, 71%, and 21%, respectively. However, 19% increase in NO_x exhaust emission was observed. Engine performance improved and specific fuel consumption (SFC) was observed to be minimal at 30% kerosene blending and decreased by 3.7% as compared to pure diesel value at the same brake power output. SFC was also observed to be minimal at 20% LPG mix and decreased by about 20% as compared to pure diesel value at the same brake power output. The fuel operating cost also reduced by 3.6% at 30% kerosene blend and further reduced by 9.6% at 20% LPG mixing with air

CHAPTER 3

SCOPE OF THE PRESENT WORK

The main objectives include :

1. Feasibility study on the production and utilization of Biofuels in automotive sector.
2. Optimize process parameters such as amount of sodium hydroxide and methanol by preparing samples of minimal quantity to identify the maximum yield.
3. To prepare biodiesel from waste cooking oil (**Palm oil**) using the proportions identified .
4. Testing the chemical properties of the biodiesel produced.
5. Engine performance testing of different prepared biodiesel blends and comparison of performance characteristics.
6. Calculation of Brake Power(BP), Brake Thermal Efficiency, Fuel Consumption and Brake specific fuel consumption(BSFC).

4.1 Preparation of Biodiesel

Biodiesel is prepared using the traditional method of *transesterification*.

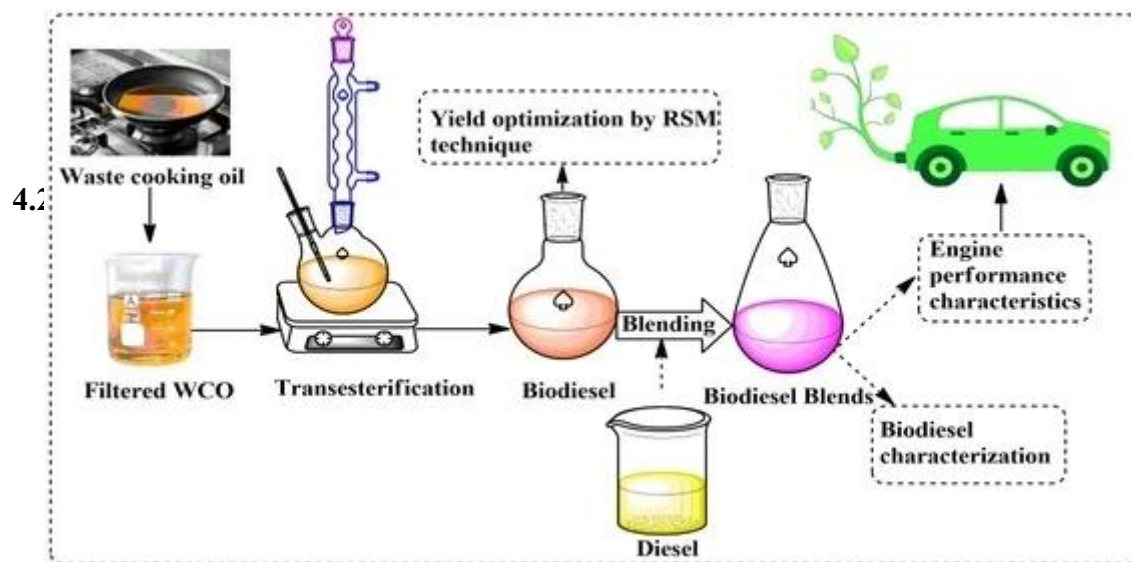


Fig. 2: Process of Biodiesel preparation

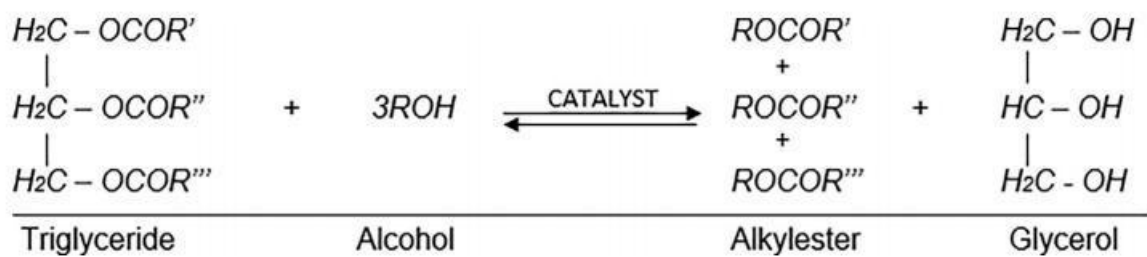


Fig. 3: Transesterification Reaction

4.2 Procedure used for Biodiesel Production:

4.2.1 Filtration of waste cooking oil : The raw waste cooking oil (Palm Oil) was taken and filtered using a cloth to remove the impurities present. The filtered oil was stored in a large glass bottle.



Fig. 4: Filtration and storing of waste cooking oil

4.2.2 Preparation of Samples : A total of 4 samples were produced using different proportions of waste cooking oil(*Palm oil*) and *Methanol* using the process of Transesterification using *Sodium hydroxide* as a catalyst.

Sample 1- Palm Oil(40ml)+ Methanol(60ml) + Catalyst **NaOH** (5mg) mixed in distilled water

Sample 2- Palm Oil(50ml)+ Methanol(50ml) + Catalyst **NaOH** (5mg) mixed in distilled water

Sample 3- Palm Oil(60ml)+ Methanol(40ml) + Catalyst **NaOH** (5mg) mixed in distilled water

Sample 4- Palm Oil(70ml)+ Methanol(30ml) + Catalyst **NaOH** (5mg) mixed in distilled water

The oil was heated to a temperature of 55-60 °C. Sodium hydroxide was first mixed with methanol together in one container before adding to the heated waste cooking oil. Each sample was taken and stirred until they became murky. Then they were heated under 60°C for about an hour while stirring constantly. The mixtures were then left to settle for 24hours.

24 hours later : Among the four samples , Sample 2 and 3 showed maximum yield of Biodiesel, *after inspection Sample 3 was selected as our final sample for further production in a larger amount.*



Fig. 5: Preparation of samples



Fig. 6: Heating of Mixture



Fig. 7: Sample 3 and Sample 2

4.2.3 Separation of Biodiesel: After 24 hours two clear layers can be seen; the top part was composed of free fatty acid methyl esters (**BIODIESEL**). The bottom deposit was made up of glycerol, salts, soap and other impurities. The contents of the beaker were transferred into a separating funnel and allowed to settle. After some time they were separated and kept in two bottles for further testing.



Fig. 8: Separating Funnel



Fig. 9: Glycerol and Biodiesel (left to right)

4.3 Testing :

4.3.1 Property Testing : The Calorific value of the prepared biodiesel was tested using Bomb Calorimeter and the Specific Gravity was calculated using a hydrometer. The Cetane Number of the sample was calculated in a lab.

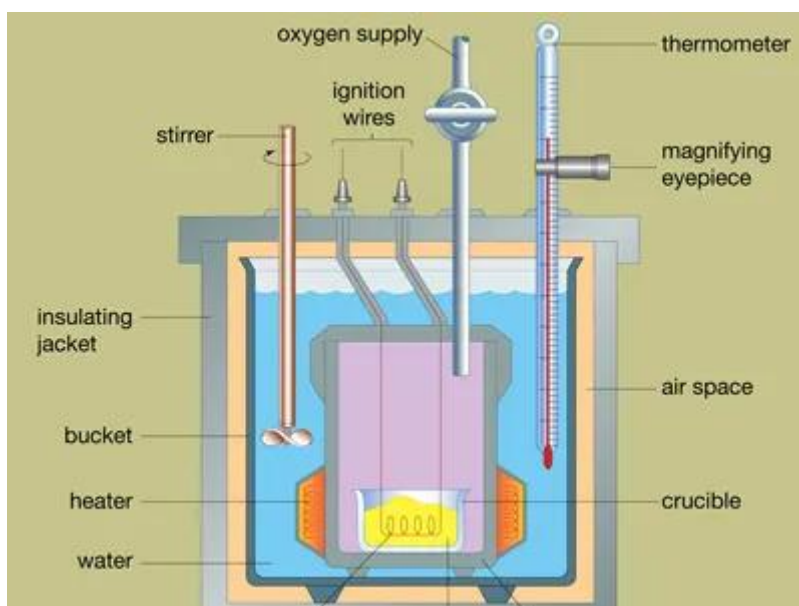


Fig. 10: Bomb Calorimeter



Fig. 11: Hydrometer

Calorific Value of Sample - 8145 kcal/kg = **34078 kJ/kg**

Cetane Value of Sample- **49**

Specific Gravity of Sample- **0.8610**

** Test Report has been added at the end for reference.*

4.3.2 Blending with Diesel : The biodiesel sample was mixed with diesel in different proportions to prepare three different test samples.

Sample 1- **B05** - Diesel((190ml) + Biodiesel(10ml)

Sample 2- **B10** - Diesel((180ml) + Biodiesel(20ml)

Sample 3- **B15** - Diesel((170ml) + Biodiesel(30ml)



Fig. 12 : Samples B15 , B10, B05 (from left to right)

4.3.3 Engine Performance Test : The test samples were fed to the engine and performance was checked at different RPM under multiple load conditions.

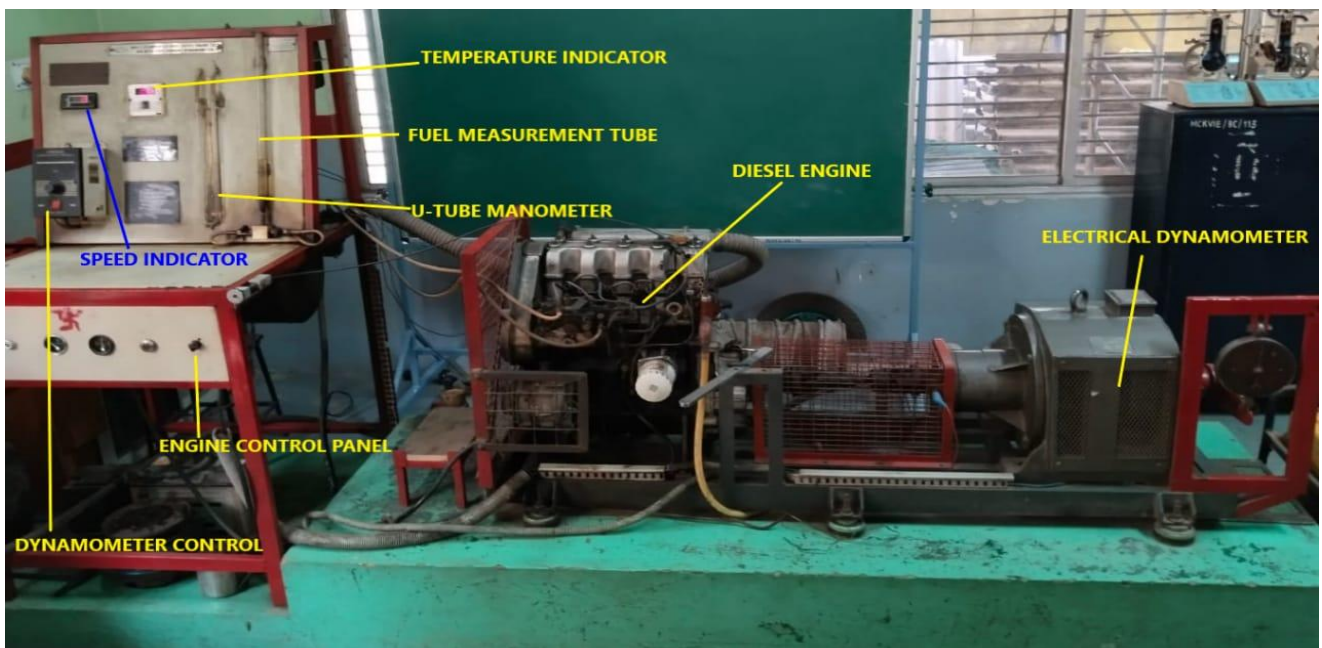


Fig. 13: Engine Test Setup

Engine Specification:

4Cylinder 4Stroke Diesel Engine.

Bore-75mm, Stroke- 79mm.

Power- 64bhp@5000rpm.

Displacement-1396cc.

Loading : Eddy Current Dynamometer (Arm Length = 200mm).

Formulae Used :

1.Brake Power (BP) = $(2\pi NT / 60000)$ kW

where N= RPM

$T = W \times 9.81 \times A_L$

where W=Load,

$A_L = \text{Arm Length} = 200\text{mm} = 0.2\text{m}$

2.Mass of Fuel Consumed (M_f) = $(10 \times \text{Specific Gravity of diesel} \times 3600) / (\text{Time} \times 1000)$ kg/hr

3.Brake Specific Fuel Consumption (BSFC) = M_f / BP kg/kw-hr

4.Brake Thermal Efficiency () = $[(\text{BP} \times 3600) / (M_f \times \text{CV})] \times 100$ %

where CV= Calorific Value of Fuel mixture

CHAPTER 5

RESULTS AND DISCUSSION

5.1. Engine Test Results:

For B05 : (5% Biodiesel + 95% Normal Diesel)

SL	Units	1	2	3	4
Load	kg	16	19	19	22
RPM	rpm	1500	1300	1700	1500
10 cc Fuel flow time	sec	20.44	19.66	15.13	15.63
Exhaust Temperature	°C	208	228	251	275
Specific fuel consumption	kg/kW-hr	0.3012	0.3043	0.3024	0.2865
Brake Power	kW	4.933	5.076	6.639	6.782
Brake Thermal Efficiency	%	27.46	27.18	27.36	28.87

For B10 : (10% Biodiesel + 90% Normal Diesel)

SL	Units	1	2	3	4	5
Load	kg	16	16	19	22	22
RPM	rpm	1300	1700	1500	1300	1700
10 cc Fuel flow time	sec	22.56	17.22	18.10	18.25	14.06
Exhaust Temperature	°C	223	242	258	265	286
Specific fuel consumption	kg/kW-hr	0.3153	0.3159	0.2868	0.2834	0.2813
Brake Power	kW	4.275	5.590	5.857	5.878	7.687
Brake Thermal Efficiency	%	26.54	26.49	29.18	29.52	29.74

For B15 : (15% Biodiesel + 85% Normal Diesel)

SL	Units	1	2	3	4
Load	kg	16	19	19	22
RPM	rpm	1500	1300	1700	1500
10 cc Fuel flow time	sec	20.03	20.15	15.46	15.60
Exhaust Temperature	°C	223	235	251	272
Specific fuel consumption	kg/kW-hr	0.3081	0.2976	0.2966	0.2877
Brake Power	kW	4.933	5.076	6.639	6.782
Brake Thermal Efficiency	%	27.48	28.45	28.54	29.43

5.2. Comparison of Performance:

BSFC vs BP

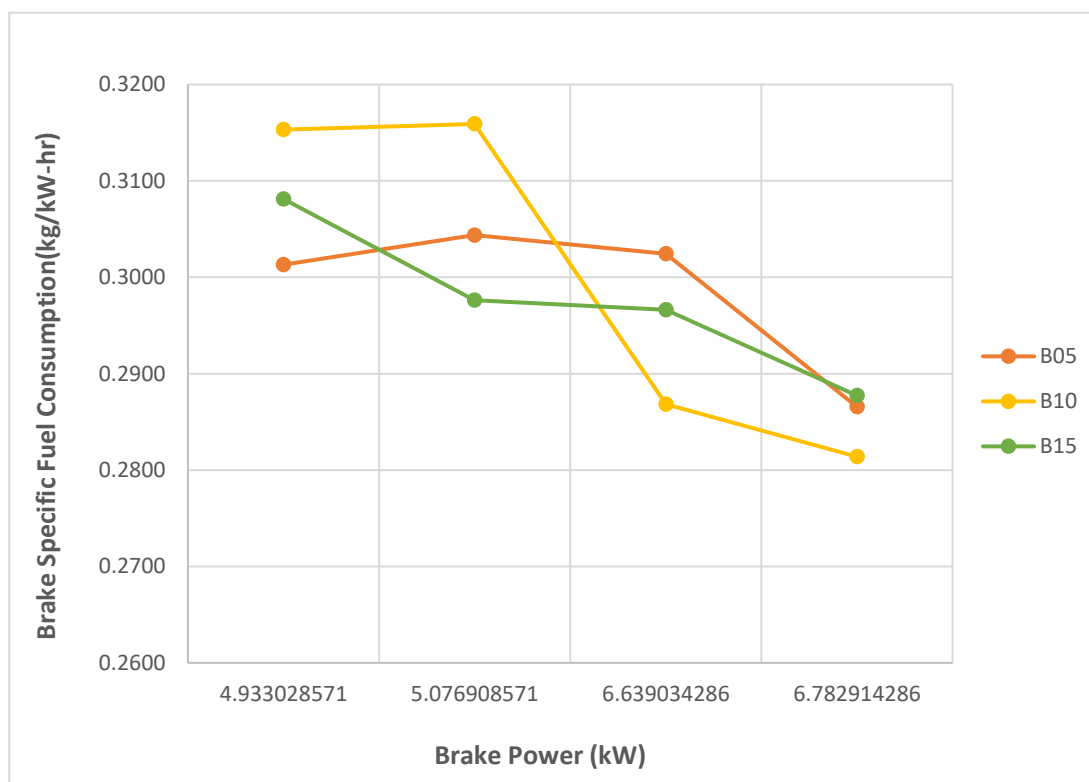


Fig. 14: BSFC vs BP

INFERENCE

- The various blends show very little variation compared to normal diesel.
- At medium load, consumption of B05 decreases and B10, B15 increases but not at higher load complete combustion of mustard oil decreases for higher blends such as B15 & B10 significantly.
- At higher load complete combustion of palm oil decreases for higher blends such as B15 and B10.

Brake Thermal Efficiency vs BP

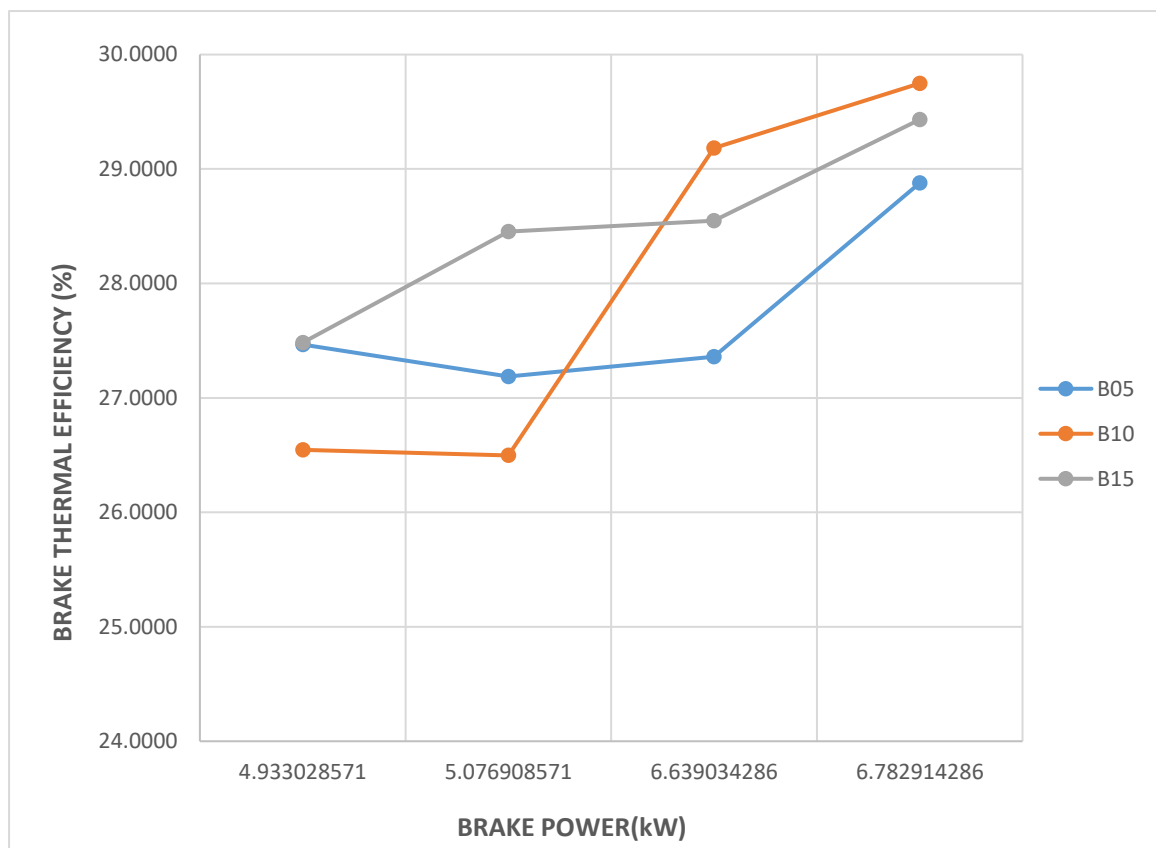


Fig. 15: Brake thermal efficiency vs BP

INFERENCE

- Brake thermal efficiency of all blends is higher than that of diesel.
- At higher loads B15 becomes more efficient than that of diesel while other blends become less efficient.

5.3. Optimization Techniques:

Entropy Method for assessing weight

There are a number of methods for assessing weight for decision-making processes such as eigenvector method, weighted least square method, entropy method etc. However, entropy method is more suitable to use when the data of the decision matrix is known. The entropy method is especially valuable to examine disparities between sets of information.

The formulation of entropy method is given in below:

Step 1: Formation of the initial decision matrix $X=[x_{ij}]_{m \times n}$

$$\text{Step 2: } E_j = -k \sum_{i=1}^n p_{ij} \ln p_{ij} \quad j=1,2,3,\dots,j \quad i=1,2,3,\dots,n \quad (1)$$

Where,

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^n x_{ij}} \quad j=1,2,3,\dots,j \quad i=1,2,3,\dots,n \quad (2)$$

and

$$k = \frac{1}{\ln n} \quad (3)$$

Where p_{ij} is a discrete probability distribution of the i^{th} alternative with respect to the j^{th} attribute.

Constant k used to ensure that $0 \leq e_j \leq 1$.

Degree of divergence, d_j can be calculated as:

$$d_j = 1 - E_j \quad j=1,2,3,\dots,j \quad (4)$$

Step 3: Final relative weights for j_{th} attribute can be obtained by simple additive normalization:

$$w_j = \frac{d_j}{\sum_{j=1}^j d_j} \quad j=1,2,3,\dots,j \quad (5)$$

Weight has been solved by the above method and mentioned below table:

Table: 1

	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)
Wi	0.127565982	0.1349733	0.737460718

MABAC method:

This phase introduces the mathematical system of the MABAC method, along with 6 steps:

Step 1. Formation of the initial decision matrix $X=[x_{ij}]_{m \times n}$. (6)

Step 2. Normalize the decision matrix as $N=[n_{ij}]_{m \times n}$.

The elements of the normalized matrix (N) are determined using the equation:

(a) For beneficial type criteria,

$$n_{ij} = \frac{x_{ij} - x_i^-}{x_i^+ - x_i^-} \quad (7)$$

(b) For non-beneficial type criteria

$$n_{ij} = \frac{x_{ij} - x_i^+}{x_i^- - x_i^+} \quad (8)$$

Where, $x_i^+ = \max(x_1, x_2, \dots, x_m)$ and $x_i^- = \min(x_1, x_2, \dots, x_m)$

Step3. Multiply the normalized decision matrix by criteria weights to determine weighted normalized fuzzy decision matrix. The same is defined as:

$$v_{ij} = w_i \cdot (n_{ij} + 1) \quad (9)$$

Step 4. The border approximation area (BAA) for each criterion is determined by the following equation.

$$g_i = \left(\prod_{j=1}^m v_{ij} \right)^{1/m} \quad (10)$$

Where, m is the total number of alternatives.

Step 5. Calculate the distance of the alternative from the BAA for the matrix elements as,

$$Q = V - G. \quad (11)$$

Step 6. Compute overall score of the alternatives using the following equation.

$$S_i = \sum_{j=1}^n q_{ij}. \quad (12)$$

Step 7. Final ranking of the alternatives are obtained by

$$S = \frac{\sum_{j=1}^n S_{ij}}{n} \quad (13)$$

Table 2:Initial decision matrix:

Alternate No.	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)
1	0.315901702	26.4974	242
2	0.302447256	27.36054	251
3	0.286561724	28.87727	275
4	0.283481756	29.52774	265
5	0.302447256	28.54891	251
6	0.31532005	27.48336	223
7	0.304376159	27.18715	228
8	0.286833969	29.18265	258
9	0.287725202	29.43175	272
10	0.297607888	28.45441	235
11	0.301299885	27.46473	208
12	0.281382507	29.74803	286
13	0.308123095	26.54628	223

Table 3: Normalize the decision matrix

Alternate No.	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)
1	0	0	0.435897
2	0.389767088	0.26553	0.551282
3	0.849961247	0.732126	0.858974
4	0.939186013	0.932231	0.730769
5	0.389767088	0.631113	0.551282
6	0.016850105	0.303312	0.192308
7	0.333887946	0.212189	0.25641
8	0.842074471	0.82607	0.641026
9	0.81625598	0.902703	0.820513
10	0.529960599	0.60204	0.346154
11	0.423005726	0.297582	0
12	1	1	1
13	0.225341497	0.015037	0.192308

Table 4: **Weighted normalized fuzzy decision matrix**

Alternate No.	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)
1	0.127565982	0.134973	1.058918
2	0.177287003	0.170813	1.14401
3	0.235992123	0.233791	1.370921
4	0.247374168	0.2608	1.276374
5	0.177287003	0.220157	1.14401
6	0.129715482	0.175912	0.87928
7	0.170158726	0.163613	0.926553
8	0.234986039	0.246471	1.210192
9	0.231692478	0.256814	1.342557
10	0.195170926	0.216233	0.992736
11	0.181527123	0.175139	0.737461
12	0.255131964	0.269947	1.474921
13	0.156311892	0.137003	0.87928

Table 5: **The border approximation area (BAA)**

	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)
BAA	0.189175853	0.199436	1.089502

Table 6: **Distance of the alternative from the BAA**

Alternative no.	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)
1	-0.06161	-0.06446	-0.03058
2	-0.01189	-0.02862	0.054507
3	0.046816	0.034355	0.281418
4	0.058198	0.061364	0.186872
5	-0.01189	0.020721	0.054507
6	-0.05946	-0.02352	-0.21022
7	-0.01902	-0.03582	-0.16295
8	0.04581	0.047035	0.12069
9	0.042517	0.057378	0.253055
10	0.005995	0.016797	-0.09677
11	-0.00765	-0.0243	-0.35204
12	0.065956	0.070511	0.385419
13	-0.03286	-0.06243	-0.21022

Table 7: Compute overall score and final ranking

Alternative no.	Qi	Final Rank
1	-0.156656682	9
2	0.013995358	7
3	0.362589522	2
4	0.30643418	4
5	0.063339359	6
6	-0.293206007	11
7	-0.217788836	10
8	0.213534804	5
9	0.352949389	3
10	-0.073974733	8
11	-0.38398714	13
12	0.521886082	1
13	-0.305519107	12

Table 8: Final Table

Input				Output			Results	
Alternative	Biodiesel Blend(%)	Load, W (kg)	RPM (N)	BSFC(Kg/KW hr)	Brake Thermal Efficiency (%)	Exhaust Gas Temperature (degree C)	Qi	Final Rank
1	10	16	1300	0.31532005	29.18265	258	-0.156656682	9
2	10	16	1700	0.315901702	26.4974	242	0.013995358	7
3	5	19	1700	0.302447256	27.36054	251	0.362589522	2
4	5	22	1500	0.286561724	28.87727	275	0.30643418	4
5	10	22	1300	0.283481756	29.52774	265	0.063339359	6
6	15	19	1700	0.302447256	28.54891	251	-0.293206007	11
7	15	16	1500	0.31532005	27.48336	223	-0.217788836	10
8	5	19	1300	0.304376159	27.18715	228	0.213534804	5
9	10	19	1500	0.286833969	29.18265	258	0.352949389	3
10	15	22	1500	0.287725202	29.43175	272	-0.073974733	8
11	15	19	1300	0.297607888	28.45441	235	-0.38398714	13
12	5	16	1500	0.301299885	27.46473	208	0.521886082	1
13	10	22	1700	0.281382507	29.74803	286	-0.305519107	12

CHAPTER 6

CONCLUSION

According to the MCDM (MABAC) optimization method, the engine yields the best results and highest Q_i at 5% blend, a 16kg load and 1500rpm. Therefore, this operation state has been designated as rank one.

- The tests demonstrate that biodiesel can easily replace conventional diesel when used in a 15% blend without requiring any engine modifications.
- Biodiesel properties improve on high loads
- The 15% blended biodiesel decreases heat input compared to normal diesel.
- Fuel consumption remains nearly identical to normal diesel under heavy loads.
- Biodiesel can minimize heat loss from exhaust gas.

From the very beginning, the goal of this project has not only been to produce biodiesel but also to raise government awareness about its benefits, encouraging the subsidization of biodiesel as a conventional fuel for all types of automobiles. While the use of 100% biodiesel in a diesel engine remains theoretical and requires long-term engine modifications, the ultimate objective is to gradually transition toward vegetable oil-based fuels. If we do not reduce our reliance on petroleum-based fuels, terms like "diesel" and "petrol" will be alien to coming generations within fifty years

CHAPTER 7

FUTURE SCOPE OF WORK

- Use an exhaust gas analyser to calculate the values of Hydrocarbon (HC), Carbon Monoxide (CO) and Nitrogen Oxides (NO_x) and check for reduction in emissions as compared to normal diesel.
- Extraction of Biodiesel from other sources such as mustard, Jatropha , Olive and perform a comparative study between them.
- Use of alternate reactants and catalyst such as hexane and enzyme treatment instead of NaOH as a catalyst.
- Perform post-treatment of glycerol to make it useful as a raw product for the glycerine industry.

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TEST REPORT

GOVT. REGISTERED

REPORT NO.: ADV/NN/2023-24/043 DATE : 24.04.2023

TEST REPORT

Customer Name and Address:	Dipesh Maiti, Souhardik Kundu, Pratyay Ghosh & Lytza Chatterjee Address : M.C.K.V. Institute of Engineering 243, G.T. Road, Liluah, Howrah, West Bengal- 711204
Sample Description	Bio-Diesel Oil
Sample Submission Date	13.04.2023
Date of Start of Analysis	17.04.2023
Date of Completion of Analysis	24.04.2023

TEST FINDINGS

SL. No	Test Parameters	Test Method	Unit	Result	Requirements
1	Gross Calorific Value	IS : 1448 (PART -6)-1984	kCal/kg	8145	Not Applicable
2	Cetane Number	IS : 1448 (PART -9)-1960	---	49	Not Applicable

For Advanced Analytical Testing Laboratory


Authorized Signatory

- The test report shall not be reproduced, except in full, without written approval of the company.
- Results related only to the parameters tested.